

Table 1: LLM Benchmark Results — Water Domain Agent (5 Models Tested)

Model	Avg Score	Speed	Quality Assessment
Llama 3.3 70B (Groq)	2.66	1.2s	Best overall: highest quality + fastest cloud response. No hallucinations, handles domain math & ISO standards
Llama 4 Scout 17B (Groq)	2.26	1.3s	Solid performer, good budget option at \$0.51/month. Occasional inconsistencies across question types
Mistral 7B (Local)	2.27	170s	Decent quality but unusably slow for real-time (nearly 3 min/query). Local privacy option only
Llama 3.1 8B (Groq)	1.68	0.9s	Fastest model but frequently hallucinates numerical values and misranks data — unreliable for production
Qwen3 32B (Groq)	1.58	2.1s	Overthinks with verbose <think> tags, inconsistent scoring. Poor value at \$1.01/month

Key Insight: Llama 3.3 70B is the clear winner — highest quality (2.66) with the fastest practical response time (1.2s). It consistently handles water domain reasoning, tariff math, and ISO 24512 compliance without hallucination. At \$1.63/month for 1,500 queries, it’s effectively free on Groq’s tier.

Note: Qwen2.5 3B (Local) was excluded from this comparison. Despite scoring 2.68 on automated metrics, it frequently hallucinates numerical values, miscalculates basic arithmetic (e.g., division errors in daily consumption), and produces unreliable rankings. Its 60-second response time also makes it unsuitable for interactive use. Quality scores alone do not capture reliability — Llama 3.3 70B is the trusted production choice.